THE ASSOCIATION OF MATHEMATICS TEACHERS OF INDIA

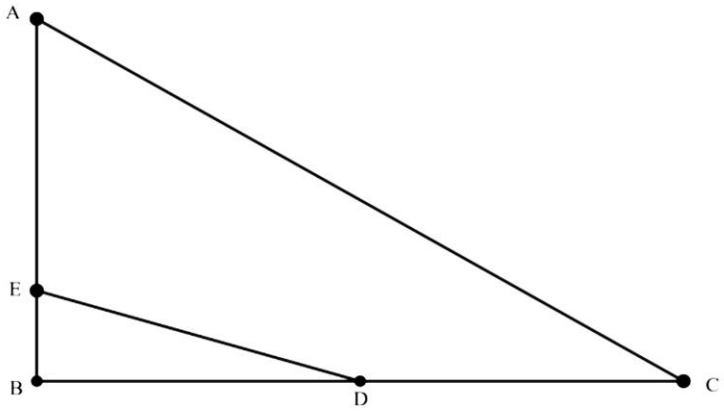
58th NMTC - SCREENING TEST - KAPREKAR CONTEST

SUB-JUNIOR LEVEL - VII & VIII GRADES

PART-A

Note: Only one of the choices A, B, C, D is correct for each question. Shade the alphabet of your choice in the response sheet. If you have any doubt in the method of answering, seek the guidance of the supervisor.

1. Let ABC be a right-angled triangle with ∠ABC = 90°. BD = 2a + 2, DC = 3a – 3, BE = 2a + 1 and EA = 4a + 1, where D is the mid-point of side BC and E is a point on side AB. The area of the semicircle with diameter AC is

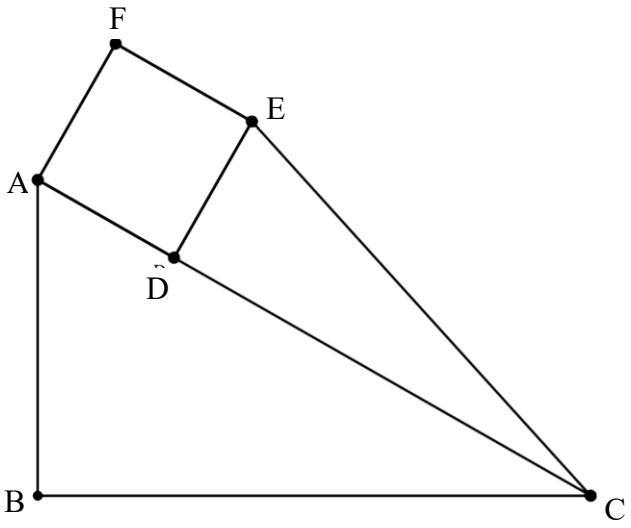


(A) 150π (B) 400π (C) 200π (D) 300π

2. If $\frac{x^2-x+1}{x^2+x+1}=\frac{11}{13}$, then the value of $x+\frac{1}{x}$ is

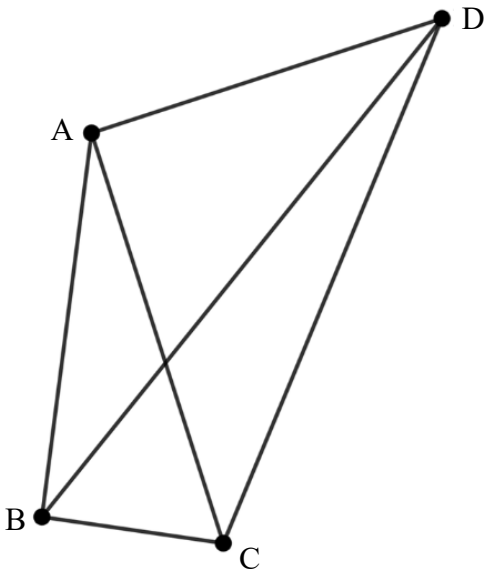
(A) 4 (B) 3 (C) 12 (D) 13

3. In the adjoining figure, ABC is a right-angled triangle, ∠ABC = 90°, BC = 12cm, area of ΔABC = 30cm^2, Area of square ADEF = 9cm^2. Then area of ΔDEC = ……cm^2.



(A) 9 (B) 16 (C) 15 (D) 30

4. ABCD is a quadrilateral in which $\angle BAC = 20°$, $\angle CAD = 60°$, $\angle ADB = 50°$, $\angle BDC = 10°$, then $\angle ACB =$



(A) 70° (B) 80° (C) 75° (D) 72°

5. If $x^2 + x + 1 = 0$, then the value of $x^{49} + x^{50} + x^{51} + x^{52} + x^{53}$ is

(A) 2 (B) 1 (C) 0 (D) –1

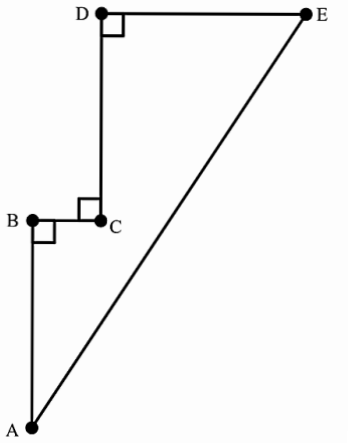
6. If $7\left(a+\frac{1}{a}\right) = \sqrt{98}$, then the numerical value of $a^5 + \frac{1}{a^5}$ is

(A) $-\sqrt{2}$ (B) $\sqrt{2}$ (C) $5\sqrt{2}$ (D) $-5\sqrt{2}$

7. If $\frac{a}{c}=\frac{b}{d}=\frac{3}{4}$ and $\sqrt{a^2+c^2}-\sqrt{b^2+d^2} = 25$, then the value of c – d is

(A) –1 (B) 7 (C) 18 (D) 20

8. In the adjoining figure AB = 5cm, BC = 1cm, CD = DE = 4cm. Then AE = ……. cm.



9. Length of one diagonal of a rhombus is 24cm. The lengths of the other diagonal and its side are both integers. How many such rhombus are possible?

(A) 2 (B) 3 (C) 4 (D) 5

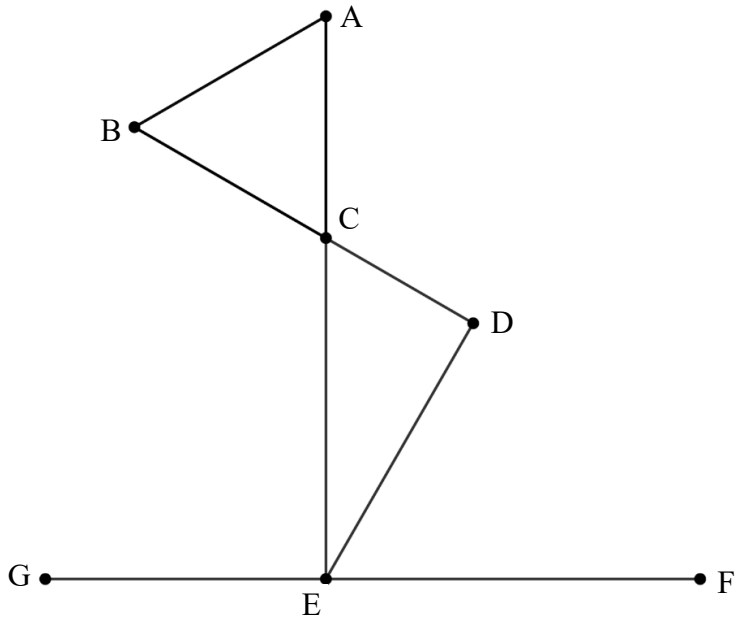
10. Both parallel sides of a trapezium are increased by 10% while its height kept constant. By what percent does its area increase?

(A) 10% (B) 21% (C) 20% (D) 11%

11. If a and b are integers with $a + \frac{1}{b} = \frac{7}{2}$ and $ab + \frac{1}{ab} = \frac{37}{6}$, then $\frac{a^2+b^2}{a^2-b^2} =$

(A) $\frac{13}{5}$ (B) $\frac{10}{3}$ (C) $\frac{8}{5}$ (D) $\frac{64}{9}$

12. In the adjoining figure, ABC is an equilateral triangle and CDE is a right-angled triangle. BCD, ACE and GEF are straight lines. ACE is perpendicular to GEF. ∠DEF = x + y, ∠GEA = 5x – y. Then the numerical value of 4x + 3y is



(A) 150 (B) 120 (C) 205 (D) 210

13. If x + y < y and x – y > –x, then the most appropriate order relation between x and y is

(A) x < 0, y < 0 (B) x < y < 0 (C) y < x < 0 (D) x < 0, y > 0

14. Simplify :

$$\frac{\left[\left(40\frac{7}{30}-38\frac{5}{12}\right)\div 10.9+\left(\frac{7}{8}-\frac{7}{30}\right)\times 1\frac{9}{11}\right]\times 4.2}{0.008}$$

(A) 100 (B) 700 (C) 1 (D) 125

15. If $a + b = 1$ and $a \times b = 2$, then $a^3 + b^3 =$

(A) –5 (B) 5 (C) –3 (D) 3

PART-B

Note: Write the correct answer in the space provided in the response sheet.

16. I have a digital clock, which displays the time, for example, as shown in the picture aside. One day afternoon, I saw the time, not in the clock but in the mirror. I realised that the mirror image showed the time three and half hours ahead of the actual time. Then the actual time was _____.



17. Isosceles trapezoid ABCD has AB = 10, CD = 20, BC = AD, and an area of 180. Then the length of BC = _____.

18. Given a regular 90-gon, we construct 90 isosceles triangles on the exterior of the polygon such that each isosceles triangle has an edge of the polygon as its base and has legs formed by the extensions of the two adjacent sides. Measure of the largest angle of one such triangle is _____ degrees.

19. The number of integers from 2 to 100 can be written as $p \times q$, where p and q are different prime numbers, is _____.

20. A positive number is mistakenly divided by 5 instead of being multiplied by 5. With respect to the correct answer, the percentage error is _____.

21. If

$$\sqrt{2\frac{23}{49}+73\frac{23}{49}}-\sqrt{4\frac{29}{49}+1\frac{15}{49}}=\frac{a}{b}$$

where a and b are coprime natural numbers, then $a + b =$ _____.

22. If $f(n+1)=\frac{3\times f(n)+2}{3}$, for all natural numbers $n > 1$ and $f(1) = 2$, then $f(2026) =$ _____.

23. Length and breadth of a given rectangle are 2026cm and 1947cm respectively. A new rectangle is formed by increasing the length and breadth of the given rectangle by x% each such that the area of the new rectangle would be double the area of the given rectangle. Then x = _____.

24. If

$$\frac{\left[\left(4.625-\frac{13}{18}\times\frac{9}{26}\right)\div\frac{x}{4}+(2.5\div1.25)\div6.75\right]\div1\frac{53}{68}}{\left(\frac{1}{2}-0.375\right)\div0.125+\left(\frac{5}{6}-\frac{7}{12}\right)\div(0.358-1.4796\div13.7)}=\frac{17}{27}$$

Then x = _____.

25. There are two vessels of equal capacity, one full of pure milk and the second one-third full of water. The second vessel is then filled up out of the first, the contents of this second are then poured back into the first till it is full. Now the ratio of quantities of milk and water in the first vessel is a:b, where a and b are coprime natural numbers, then a + b = _____.

26. If $\frac{x-1}{n-1}+\frac{2n^2(1-x)}{n^4-1}=\frac{2x-1}{1-n^4}-\frac{1-x}{1+n}$, then x = _____.

27. n is a positive integer such that 12n is divisible by 12 + n, then the number of possible values of n is _____.

28. Consider the sequence 2, 2, 4, 4, 4, 4, 6, 6, 6, 6, 6, 6, … . Then its 100th term is _____.

29. Let P(n) and S(n) denote the product and the sum, respectively, of the digits of the integer n. For example, P(23) = 6 and S(23) = 5. Suppose N is a two-digit number such that N = P(N) + 3S(N). Then N = _____.

30. Area of a triangle with side lengths 17cm, 25cm, 26cm respectively, is _____cm^2.

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